Group 20

Web-Based Hospital Medicine Ordering and Line-Following Autonomous Delivery Rover

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PROBLEM & MOTIVATION

Problem

- Hospital staff manually deliver medicines to patient rooms
- Time-consuming and inefficient
- Increased workload and chances of delay or error

Motivation

- Automate internal hospital logistics
- Reduce unnecessary human movement
- Improve delivery speed and safety

Objectives of the Project

1. To design a responsive web portal for staff order management.
2. To implement a secure cloud-based backend using Supabase.
3. To develop an autonomous robot capable of line-following and junction-based indoor navigation.
4. To ensure real-time status synchronization between the robot and the web portal.

PROPOSED SOLUTION

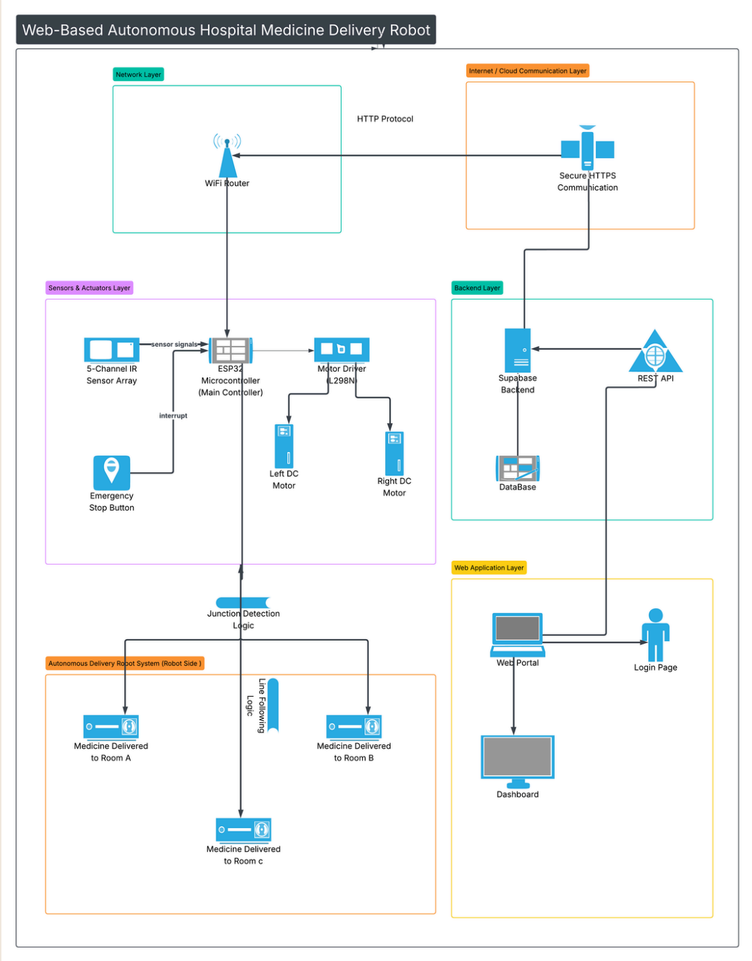
Proposed System

- A web application is provided to hospital staff to place medicine delivery requests.
- The backend system processes the request and selects a predefined delivery route.
- An autonomous line-following rover receives the route instructions and performs the delivery.

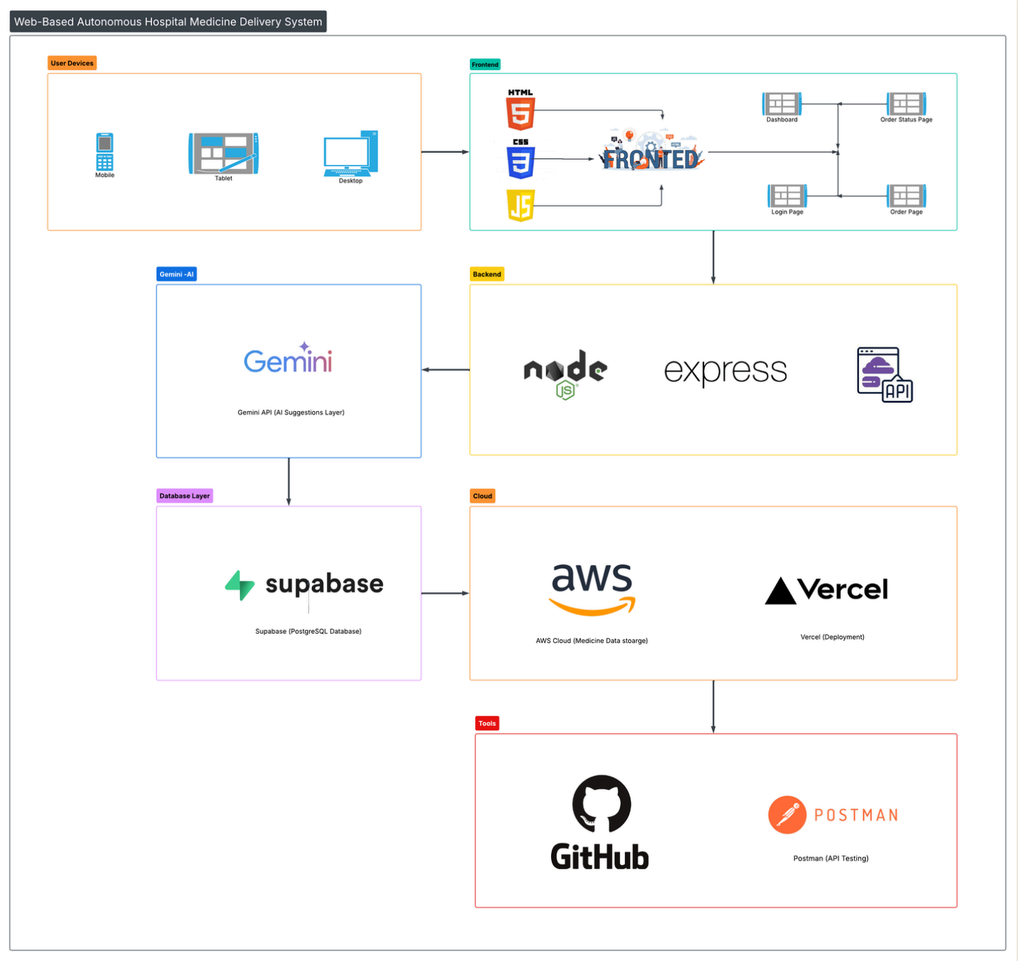
Key Concept

- High-level decisions are made by the backend software. Low-level movement and navigation are handled by the rover.

System Architecture



Total System Architecture (Soft+Hard)



Software Architecture

Hardware Specifications

- **Microcontroller**: ESP32 (Dual-core, Wi-Fi enabled for DB communication).
- **Navigation**: 5-Channel IR Sensor Array (High-precision line following and junction detection).
- **Safety**: Ultrasonic Sensors (HC-SR04) for obstacle avoidance.
- **Mobility**: 2WD Chassis with L298N Motor Driver and high-torque DC motors.
- **Power**: 12V Li-ion Battery pack with a buck converter for ESP32 power.

Software Stack & Methodology

- **Frontend**: HTML5, CSS3, Bootstrap 5 (for responsive layout), and JavaScript.
- **Backend**: Supabase (PostgreSQL for data, Auth for staff login, and Realtime for live tracking).
- **Firmware**: C++ using the Arduino Framework for ESP32 control logic.
- **Methodology**: * Modular Design: Separate development of the Web UI and the Robot Firmware.
- **State Machine Logic**: The robot operates in states: IDLE, FETCHING, MOVING, OBSTACLE_DETECTED, DELIVERED.

LCA-1

Web Portal Interface (Completed)

We successfully developed the core user journey using HTML, CSS, Bootstrap, and JS:

1. Home Page: Featuring a dynamic medicine search bar.
2. Medicine Details: Page displaying descriptions and dosages.
3. Cart System: For managing selected medicines.
4. Checkout Page: Interface to input destination room numbers for delivery.

Hardware & Research

- Component Selection: Finalized all sensors and motor specs.
- Brainstorming: Completed junction-routing logic design.
- Research: In-depth study on hospital corridor navigation and Wi-Fi dead-zone handling.

Current Progress (LCA 2 Status)

- Software Development (Upgraded): Transitioned from a static HTML/Bootstrap UI to a dynamic, interactive frontend using React.js. Implemented full State Management for the medicine search and cart systems.
- Hardware Milestone (Achieved): Completed the full physical assembly of the MedRover. Successfully implemented and tested Line-Following Logic using the 5-channel IR sensor array
- Full-Stack Connectivity: Established real-time communication between the React frontend and the Supabase Backend using Node.js.

Plans for LCA-3 (The Next Phase)

- Cyber-Physical Integration (The Handshake): Establishing real-time synchronization between ESP32 firmware and Supabase to enable instant command delivery and automated status updates on the React dashboard.
- AI-Driven Clinical Support: Implementing an AI Tablet Suggestion System to provide intelligent medicine substitutes and safety-critical drug-drug interaction checks for enhanced clinical decision-making.
- Autonomous Navigation Optimization: Refining junction-based logic and IR sensor precision to ensure the rover executes complex multi-room routes and matches digital pathing instructions accurately.

Conclusion & Future Scope

Conclusion:

- Our project addresses the critical need for automation in healthcare logistics. By offloading routine delivery tasks to an autonomous system, we reduce the workload on nursing staff, minimize human error in medicine distribution, and ensure a sterile, contact-free delivery process—essential in modern post-pandemic hospital environments.

Future Scope

Future Scope

- Cloud Scaling: Transitioning from Supabase to a full AWS deployment using AWS IoT Core for industrial-grade fleet management and high-scale reliability.
- Admin Analytics: A comprehensive dashboard to visualize daily performance, including delivery success rates and operational metrics.
- Patient Tracking: A dedicated module to maintain historical delivery logs mapped to specific patients for improved record accuracy.
- Multi-Robot Coordination: Developing communication protocols to allow multiple robots to navigate the same corridors without collisions.